

SCANNING AND TRANSFER OF CRYOGENIC SAMPLES IN THE Bionanoprobe-II INSTRUMENT AT THE ADVANCED PHOTON SOURCE*

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Abstract

A new hard x-ray fluorescence (XRF) nanoprobe instrument called Bionanoprobe-II (BNP-II) has been designed and will be constructed at 2-ID-D of the upgraded Advanced Photon Source. BNP-II will take advantage of the orders-of-magnitude increase in brightness and coherent flux with advanced sample scanning, metrology, cryogenics, and controls. These advancements will enable high-throughput XRF imaging under cryogenic conditions with 10 nm spatial resolution, 2D survey of mm-sized samples, and fast tomography for 3D visualization. BNP-II also introduces a novel robotic sample transfer system that interconnects a cryogenic plasma focused ion beam (cryo-PFIB) milling station alongside the x-ray nanoprobe. The interconnected instruments enable an iterative workflow between x-ray measurements and cryo-PFIB milling and maintains the integrity of vitrified samples by remaining below 110 K even during transfer. Regions of interest can be identified by fast large-area scans, after which the sample geometry can be optimized for nanoscale x-ray imaging and tomography. This work details the engineering advancements required to examine highly complex, multidimensional systems with BNP-II.

INTRODUCTION

The Bionanoprobe-II (BNP-II) at Argonne National Laboratory Advanced Photon Source (ANL APS) Sector 2-ID-D is a hard x-ray fluorescence (XRF) nanoprobe, currently under construction, that will utilize the orders-of-magnitude improvement of brightness and coherent flux of the recently upgraded APS [1]. This next-generation instrument (see Fig. 1) will have sub-10 nm resolution and will enable both fast 2D survey of mm-sized samples and fast XRF tomography for 3D visualization. The instrument is dedicated to studying trace elements in frozen-hydrated biological samples. The samples remain in a frozen-hydrated state from sample preparation stage, during transfer to the beamline, and during data collection.

The first major goal for BNP-II is to achieve 10-100x greater experimental throughput compared to the current instrument's ability [2], meaning substantially larger populations of biological cells are imaged in a shorter time.

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This faster imaging along with higher resolution requires advanced sample scanning hardware, metrology, and controls development.

The second major goal for BNP-II is to integrate a sample milling station alongside the x-ray nanoprobe. The interconnected instruments enable an iterative workflow between x-ray measurements and sample milling, where regions of interest can be identified by fast large-area scans, after which the sample can be milled down for better nanoprobe access.

The combination of high experimental throughput and micron-level targeted imaging of biological samples establishes BNP-II as a frontier capability in life and soft matter sciences.

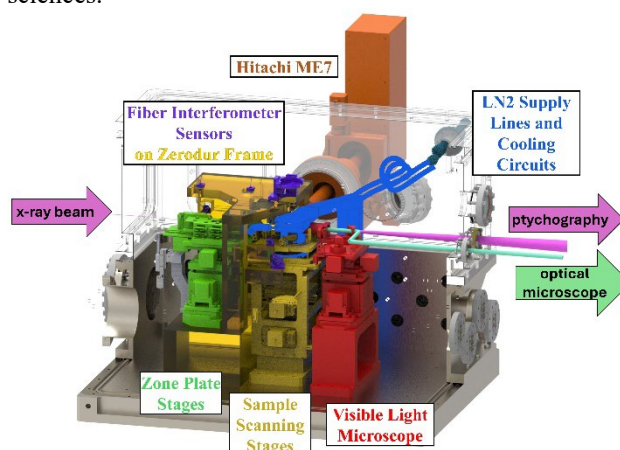


Figure 1. Overview of the BNP-II instrument core

CRYOGENIC SAMPLE TRANSFER

Effective imaging and analysis of biological samples in their 'natural state' requires keeping the samples in a hydrated state and avoiding radiation-induced damage. The integrity of a sample is preserved if it is vitrified and remains below the amorphous-to-crystalline ice transition temperature during transportation, loading, x-ray imaging, and cryo-PFIB milling. Warming of the specimen could cause a phase change and consequent structural and chemical alterations. Three new engineering advancements have been made to prevent such sample damage (see Fig. 2).

Sample Holder, Shuttle, and Dock Systems

BNP-II samples are handled and transported with a hierarchy of interfaces, enabling safe handling at a cryogenic temperature. Samples are fast frozen with liquid nitrogen (LN2) onto either TEM grids or silicon chips with silicon nitride films. The grid or chip is then mounted onto a

sample holder stick, which is inserted into a shuttle. A transport case then takes the shuttle and drops it off onto a docking station inside BNP-II (see Fig. 3). The sample holder is designed around a standard interface, to ensure repeatable and reliable mounting with good thermal contact.

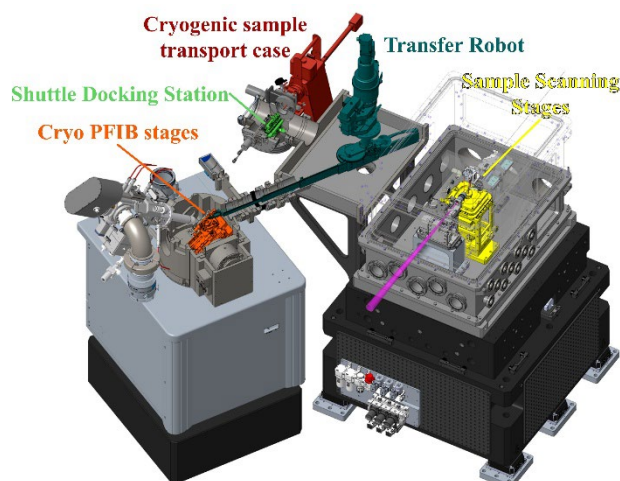


Figure 2. BNP-II robot sample transfer system. The robot picks up and drops off sample holders at three locations: docking station, x-ray sample scanning stages, and cryo-PFIB stages.

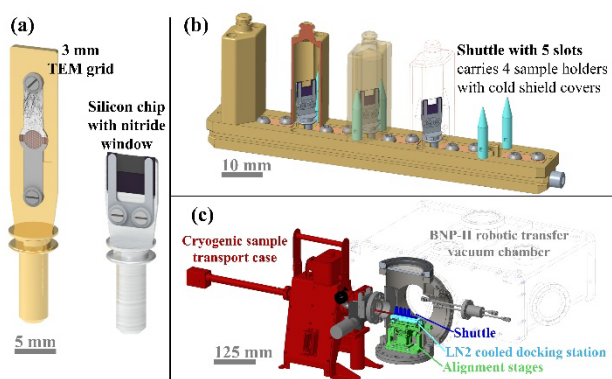


Figure 3. (a) BNP-II sample holder examples (b) 4 sample holders in a shuttle (c) A shuttle being loaded in the chamber docking station via a cryo sample transport case.

Robot and Gripper

Once the sample shuttle is loaded in the instrument docking station, a robotic sample transfer system (robot) picks up one sample at a time, and either transfers it to the BNP-II sample scanning stages or the cryo-PFIB stages. The robot can then make further transfers to either iteratively mill and image or select a different sample holder.

The robot is comprised of a commercially available rotary telescopic transfer arm (RTTA from UHV Design) with a gripper mechanism at the end (under development by APS). The RTTA provides 360° rotation, 760 mm linear extension, and 50 mm vertical motion. The gripper has copper fingertips that are pre-cooled before picking up a sample. Their thermal mass must be sufficient to prevent the sample holder from warming above the devitrification

threshold, due to combined radiative heating from the surroundings and conduction through an imperfect thermal break mechanical connection to the gripper. The gripper assembly (see Fig. 4) also includes three capacitive displacement sensors to measure relative XYZ position of a reference pillar, at each of the three locations where samples are picked up and dropped off. XYZ stages at all three locations, in a closed loop control scheme, use this live position data to accommodate misalignment of the gripper, inherent to the RTTA's long arm extension.

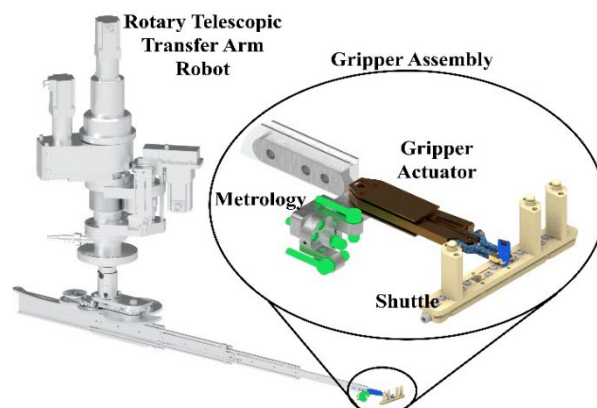


Figure 4. BNP-II sample transfer robot and gripper.

Interconnected Vacuum Chambers

Connecting the cryo-PFIB instrument (discussed further in the next section) to the BNP-II instrument chamber is not trivial. X-ray nanoprobe instruments are constructed for maximum rigidity to the floor, in order to register with the synchrotron light source. In contrast, scanning electron microscopes (SEMs) and PFIBs have the illumination source on board the vacuum chamber, and can therefore be mechanically and vibrationally isolated from the floor. Our cryo-PFIB instrument uses 4 pneumatic vibration isolators, which permit free translation by ~10 mm in XYZ, and free rotation by ~3° about XYZ. After exploratory vibration measurements and SEM/PFIB imaging tests, we determined that restricting any of the 6 degrees of freedom (DOFs) is detrimental. This necessitates connecting the chambers with a bore opening large enough for the robot arm and gripper to pass through (>70 mm), while still allowing the floating PFIB chamber to move freely, with minimal added rotational inertia or friction.

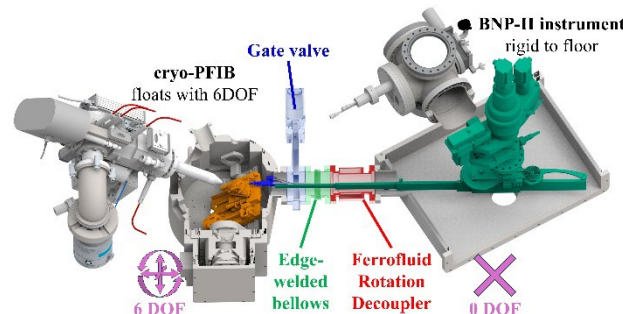


Figure 5. Connection between BNP-II instrument chamber and cryo-PFIB must allow 6 DOFs and a large bore.

We achieve this result with a combination of an edge-welded bellows, that permits 5 DOFs (all but rotation) and a newly conceived device, coined a rotational decoupler. This rotational decoupler, under development by APS and Kurt J Lesker, is constructed from a commercially available ferromagnetic fluid rotary coupling, with conflat flanges welded on both ends. To the authors' best knowledge, this device is newly conceived and therefore a patent is pending. While the robot is transferring samples, the pneumatic control will be turned off temporarily, to avoid an unstable drop-off location for the sample holder.

CRYOGENIC SAMPLE MILLING

A Zeiss 1540XB scanning electron microscope has been fitted with both a Xenon plasma FIB column developed by Tescan USA and a cryogenic sample cooling stage developed by Leica. The workflow for milling samples begins with locating a region of interest (ROI) with XRF microscopy, ptychography, or optical fluorescence microscopy system that is integrated into BNP-II. The sample is transferred to the cryo-PFIB chamber without warming, and a site-specific isolation of identified sub-volumes is performed to optimize the sample geometry [3].

SAMPLE SCANNING AND METROLOGY

The sample stage stack (see Fig. 6) is comprised firstly of a set of XYZ cross-roller guided, stepper motor / lead screw actuated stages from Kohzu, provide coarse alignment capability of 10 mm³ travel range with <1 µm repeatability, and absolute encoders of 50 nm resolution. Above the coarse stages, a Micronix PR-70 rotation stage with an integrated absolute encoder is used for 3D tomographic data collection. The wobble error of the rotation stage is measured with the interferometer metrology system and compensated with an APS-designed flexure guided, piezo actuated pitch correction stage. Just above the rotation stage, a Micronix PPX-50 stage allows centering any point of interest (POI) from the sample on rotation axis.

Sample scanning is executed with an XYZ scanning stage (PI-USA P-616.3C, customized for high vacuum) with 100 µm³ travel range and a repeatability of 10 nm. 2D scans are typically performed in a snake pattern, with continuous and fast vertical motion as the inner scan loop. Tomographic data acquire consists of a series of such 2D scans collected at different sample rotation angles.

A cryogenic sample holder interface is mounted on top of the rotation stage with a thermal break. This "cold chuck" is actively cooled to approximately 100 K with a copper strap leading to a LN₂ cooled block. A cylindrical reference mirror is integrated into the cold chuck. A travelling interferometer platform tracks this mirror in the horizontal plane and is decoupled from the other degrees of freedom of the sample stage stack [4]. In total, 15 fiber interferometer laser axes are used: 3 sensor heads on the travelling platform for tracking the cylindrical mirror, 6 sensor heads that measure horizontal motion of the traveling platform and the vertical motion of the stage stack, 4 sensor heads measure the zone plate focus position, and 2 more to measure a high-resolution zone plate focal position.

During sample scanning, all interferometer signals are recorded at defined frequencies in conjunction with the XRF spectra under a shared clock. In post-experiment data analysis, this information will be integrated to assign spectra to individual pixels, providing spatial resolution down to 10 nm.

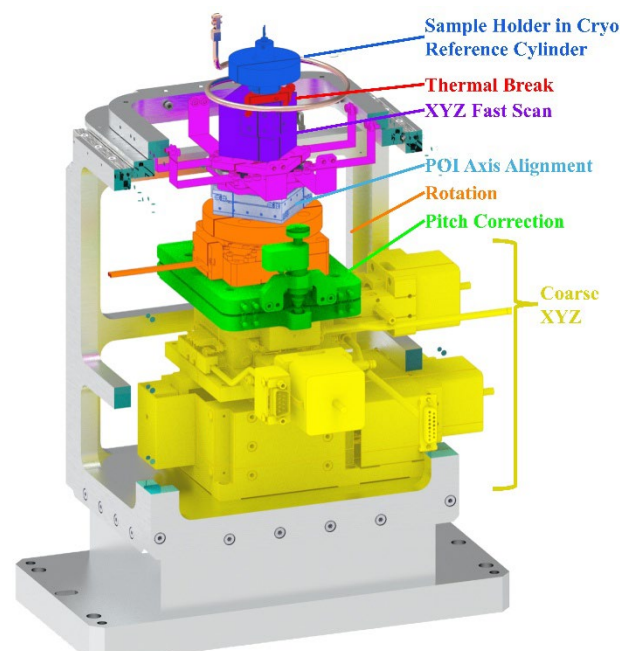


Figure 6. BNP-II Sample Scanning stages. Partial cross-section view through the travelling interferometer frame.

CONCLUSION

The BNP-II instrument combines high experimental throughput with sub-micron-level specificity for isolating a region of interest from neighboring bulk in biological cells. This combination establishes BNP-II as a frontier capability in life and soft matter sciences.

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